

The equality case in the weak triple-minimum support bound

Research note: a general Gram-matrix obstruction

30 September 2026

Consider $q = 3h$ equally weighted permutations on $n = q + 1$ labels such that each label is earliest in every triple containing it in exactly h rows. Write r_{ij} for the number of rows with $i < j$. Necessarily $h \leq r_{ij} \leq 2h$. Call a label bad if $r_{ij} = h$ for some opponent j , and good otherwise. A bad label is globally first in exactly h rows: the event $i < j$ already has the same size as every event $i < j, i < k$. Thus there are at most three bad labels, and at least $q - 2$ good labels.

Lemma 1. *For a good pivot i , put $e_j = r_{ij} - h \in \{1, \dots, h\}$ over its q opponents. Then*

$$q = \sum_j e_j + h - \frac{h(q-1)^2}{1 + h \sum_j e_j^{-1}}, \quad (1)$$

and the positive integer

$$\left(\prod_j e_j \right) \left(1 + h \sum_j e_j^{-1} \right) \quad (2)$$

is a square.

Proof. Let C be the $q \times q$ zero-one matrix whose row is a permutation and whose opponent column j indicates $i < j$. Triple-minimum uniformity gives

$$C^T C = D + h \mathbf{1} \mathbf{1}^T = G, \quad D = \text{diag}(e_j), \quad C^T \mathbf{1} = e + h \mathbf{1}.$$

Since D is positive definite, C is invertible. Consequently

$$q = (e + h \mathbf{1})^T G^{-1} (e + h \mathbf{1}).$$

Sherman–Morrison gives (??); the matrix determinant lemma gives (??) as $\det G = (\det C)^2$. $\square$

Lemma 2. *Every good label avoids the last two positions in every row. Consequently there are exactly three bad labels.*

Proof. For the matrix C of Lemma ??, solve $Cw = \mathbf{1}$. Sherman–Morrison applied to $Gw = C^T \mathbf{1} = e + h \mathbf{1}$ gives

$$w_j = 1 - \frac{\alpha}{e_j}, \quad \alpha = \frac{h(q-1)}{1 + h \sum_j e_j^{-1}} > 0.$$

Thus every $w_j < 1$. A row of C with zero or one entries equal to one cannot have scalar product one with w . The pivot therefore has at least two opponents after it in every row.

The last two positions must both be bad labels. If there were at most two bad labels, they would occupy these positions in every row and could never be first. This contradicts the fact that each bad label is first in $h > 0$ rows. There are at most three bad labels, so there are exactly three. $\square$

Lemma 3. *If there are exactly three bad labels, then every good pivot has exactly three entries $e_j = h$.*

Proof. The bad labels account for all first positions. For any good label a , sum its minimum counts over the three pairs of bad labels. The total is $3h = q$. In any row at least one bad label precedes a , so that row contributes at most one to the sum. Every row must therefore contribute one: a follows the first bad label and precedes the other two. Each bad label is first h times, hence $r_{ab} = 2h$ for every good a and bad b . Conversely, $e_j = h$ means $r_{ji} = h$, so j must be bad. $\square$

Theorem 1. *For every integer $h \geq 2$, no weak triple-minimum family of $3h$ equally weighted permutations exists on $3h + 1$ labels.*

Proof. By Lemmas ?? and ??, there are exactly three bad labels. Each row consists of its first bad label, all good labels, and the remaining two bad labels. Partition the rows into three groups of size h according to their first bad label.

Fix two distinct good labels a, c , and write $r = r_{ac}$. For each bad label b , let x_b count rows with $a < c$ in the group beginning with b . In all other rows both a, c precede b . The minimum condition on the triple $\{a, c, b\}$ gives $r - x_b = h$. Summing over the three groups gives $r = \sum_b x_b = 3(r - h)$, and hence

$$r = 3h/2, \quad e_{ac} = h/2.$$

There are $3h - 2 \geq 4$ good labels, so such a pair exists and h must be even.

Fix a good pivot. Its three bad opponents have $e_j = h$, while its other $3h - 3$ opponents have $e_j = h/2$. Formula (??) now gives

$$(\det C)^2 = \det G = h^3(h/2)^{3h-3}(1 + 3 + 2(3h - 3)) = \frac{h^{3h}(3h - 1)}{2^{3h-4}}.$$

Because h is even, both h^{3h} and 2^{3h-4} are squares. Thus $3h - 1$ must be a rational square, and therefore an integer square. This is impossible modulo three. $\square$

Support consequence. Every equally weighted weak triple-minimum family on $n \geq 5$ labels has at least n rows. Indeed, the general rank argument in `../partial_fairness/weak_triple_minimum_support.tex` gives $q \geq n - 1$ even for unequal positive weights. In the equal-weight case $3 \mid q$, and equality $q = n - 1$ would fall under the theorem. The exception $h = 1$ is real: on four labels the three permutations 1203, 3201, 0213 are weak triple-minimum uniform.

The new equality obstruction uses equal row weights, integrality of the order counts, and the square determinant of an integer matrix. It makes no stronger claim about unequal positive row weights. The earlier note `six_weak_minimum_permutations.tex` gives an independent position-based proof of the $h = 2$ exclusion. The exact small-profile enumeration in `check_weak_minimum_profiles.py` and its JSON output is retained as a computational companion; the general proof above does not depend on it.